

Second part: Open book

Total score: 10 pts.

You are warmly invited to be tidy and concise. Messy papers will be penalized.

5. A natural gas dealer has to plan the purchase of gas for the next T days. The dealer knows the forecasted gas price p_t per unit for every day $t = 1, \dots, T$. The gas is stocked in a depot. The capacity of the depot is Q , thus at any day, the depot should contain no more than Q . This capacity, if needed, can be increased of Q^+ at the cost of c_t in day $t = 1, \dots, T$. The dealer knows also the demand of the market: $d_t, t = 1, \dots, T$ and the demand must be met exactly. Initially the depot contains m_0 . In order to avoid the risk of shortages, the dealer wants to maintain at least q units of gas in the depot. However, this is a soft constraint, that is, in case an economical advantage, it is possible to reduce the presence of gas in the depot at the cost of r_t per unit. Formulate the problem of planning the purchase and stocking of the gas at the minimum cost, so that the demand is met and the capacity constraints are satisfied.

Variante: Let us assume that the purchase of the gas has a lead time of Δ . This means that if we buy a unit of gas at time t , the gas will arrive after Δ days. How does the formulation change?

6. The manager of a beach has to accept reservations for the upcoming season. Due to the sanitary emergency she has a very limited number of umbrellas (n) and she is not allowed to accept customers that do not rent umbrellas. The m customers have issued their requests. The request of each customer i is characterized by the starting day s_i , the ending day e_i and the estimated revenue r_i for having him as a customer. The manager has to select which customers to accept so as to maximize the total estimated revenue, while not exceeding the maximum number of occupied umbrellas. Note that a customer is either accepted for the whole requested period or rejected. Formulate the problem of the beach manager as a minimum cost flow problem, thus specify the set of nodes, the set of arcs, the balances of the nodes, the costs and the capacities of the arcs.